

DURGASHAMBAVI AMARNENI

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SUMMARY

Motivated Software Engineer with a strong background in database programming, software development, and debugging. Experienced in designing and implementing applications using AWS, Azure, Python, Java, SQL, PL/SQL, and Oracle ERP. Adept at solving complex problems, collaborating in agile environments, and building innovative solutions. Currently pursuing a master's in computer science at Indiana State University, seeking a full-time opportunity to leverage technical skills and contribute to cutting-edge software projects.

EDUCATION

Master of Science in Computer Science

Indiana State University | January 2024 - May 2025 | 3.91 CGPA

Relevant Coursework: Machine learning, Artificial Intelligence, Database management systems, Web programming, Cloud computing, Data Science, Data Mining

Bachelor of Technology in Computer Science Engineering

Anurag Engineering College | 2017 - 2021 | 89%

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, Golang, C, C++, XML
- **Web Technologies:** HTML, CSS, JavaScript
- **Databases:** SQL, PL/SQL, Oracle ERP
- **Cloud:** Microsoft Azure, AWS
- **Machine Learning and Artificial Intelligence**
- **DevOps & Automation:** DevOps, CI/CD Pipelines, Docker, Kubernetes
- **Software Development:** Object-Oriented Programming, Microservices Architecture, Data Structures & Algorithms, AngularJS
- **Tools:** Tableau, Power BI
- **Development Methodologies:** Agile, SDLC

WORK EXPERIENCE

Tata Consultancy Services | Aug 2021 - Jan 2024

Role: Oracle ERP Developer

Project: Tirumala Tirupathi Devasthanams (CSR-TTD)

Technologies: SQL, PL/SQL, Oracle Apps ERP, Java

- Collaborated with business analysts to gather and analyze business requirements and translate them into technical solutions.
- Designed and developed robust database structures, including tables, triggers, stored procedures, and functions using SQL and PL/SQL.
- Developed and customized Java-based Oracle Applications Framework (OAF) pages, enhancing user interaction and modular functionality within Oracle ERP.
- Integrated Java classes with PL/SQL procedures to ensure seamless backend data processing and validation.
- Implemented form personalization in the Inventory (INV) module.
- Utilized JavaBeans and BC4J components for business logic implementation and data handling in OAF pages.
- Automated deployment of OAF pages using Unix shell scripts, moving applications to the production environment.
- Created Oracle APEX applications.
- Wrote Java code for controller extensions, validations, and page-level event handling to meet specific business rules.
- Implemented Java-based web services to enable integration with third-party systems and internal modules.
- Created and optimized XML Publisher reports to meet evolving business needs and improve data presentation.
- Applied form personalization techniques in the Inventory (INV) module to extend native Oracle functionality.
- Developed SQL Loader control files and PL/SQL programs for high-performance data loading and validation.
- Automated deployment of OAF pages using Unix shell scripts, streamlining application migration to production.
- Built interactive Oracle APEX applications to support internal process automation and user-friendly data access.

Indiana State University | March 2024 - Present

Role: Computer Technician, OIT (Technology Support Center)

- Diagnosed and resolved software/hardware issues for faculty and students.
- Assisted users with troubleshooting and technical support.
- Maintained IT infrastructure and provided front-line user support.

Role: CETS AV/IT Technician (March 2024 - September 2024)

- Managed projectors, public printers, and desktop setups.
- Provided first and second-line technical support for classroom technology.
- Assisted in audio/visual setups, video engineering, and event recording.

INTERNSHIP EXPERIENCE

Software Trainee | Indian Servers | May 2019 - July 2019, May 2020 - July 2020

- Gained hands-on experience in software development.
- Worked on some of the real-time projects.

PROJECTS

- **Automatic Helmet Detection:**
 - Developed a real-time helmet detection system using Python and Machine Learning techniques, employing Convolutional Neural Networks (CNNs) for classification of helmet usage in surveillance footage.
 - Built a dual-stage machine learning pipeline: one CNN for detecting motorcyclists/persons, followed by a second CNN for helmet presence classification based on localized head regions.
 - Utilized OpenCV with Python for video frame extraction, motion detection, and adaptive background subtraction from CCTV streams.
 - Applied data preprocessing, visualization, and model evaluation using NumPy, Pandas, and Matplotlib to enhance training performance.
 - Engineered a cost-effective, Python-based AI solution leveraging existing CCTV infrastructure to reduce human intervention in safety enforcement.
 - Performed unit testing, system testing, and integration testing to ensure model robustness and accuracy across edge cases.
- **Voice-Based Email Application:**

- Developed a voice-controlled email web application in Python using the Django framework, designed to help visually impaired users send, receive, and manage emails through voice commands.
 - Applied Machine Learning-based speech technologies using Google Speech Recognition API for speech-to-text conversion and gTTS (Google Text-to-Speech) for dynamic audio responses.
 - Implemented core functionalities like secure login, inbox, compose, sent, and trash features using Python's smtplib and imaplib libraries for handling SMTP/IMAP protocols.
 - Built a responsive, accessible UI using Django templates and Bootstrap, supporting both voice-driven and manual interaction modes.
 - Developed audio-based attachment features, including recording, generating, and playing .mp3 files using Python libraries like playsound and gTTS.
 - Designed a robust voice-guided workflow with intelligent confirmation prompts and error handling, enhancing overall user experience and accessibility.
- **Website Enhancement – *Department of Computer Science, Indiana State University:***
 - Tasked with enhancing the existing official departmental website (cs.indstate.edu) to improve its UI, visual appeal, and content structure.
 - Implemented frontend updates using HTML, CSS, JavaScript, and Bootstrap, focusing on layout, accessibility, and responsiveness.
 - Collaborated with faculty and IT staff to add new content sections (events, faculty profiles, resources), improving information availability for students and visitors.
 - Worked on optimizing user experience by refining navigation, fixing styling inconsistencies, and ensuring cross-device compatibility.
 - Contributed to making the website more engaging, student-friendly, and modern-looking while maintaining academic standards.